

Frequency-Aware Spectral Conditioning for Latent Graph Diffusion

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TL;DR. We study whether low-frequency spectral information from the graph Laplacian can improve topology-only latent diffusion models for graph generation. We test whether there is a useful frequency cutoff: too little spectral information may miss global structure, while too much high-frequency information may add noise and hurt generation quality.

1 Background & Motivation

Graph generation aims to learn a distribution over graphs and generate new realistic, valid, and diverse graphs. In this project we focus on **topology-only graph generation**: every graph is represented only by its adjacency matrix A , without node features, node types, edge types, or edge features. This keeps the project focused on the structural question: how can a generator learn the connectivity pattern of a graph?

This setting is useful for synthetic graph data augmentation and topology-based GNN research. If we can generate realistic planar graphs or community-structured graphs, they can be used as additional training data when real labeled graphs are limited, or as controlled benchmarks for testing whether GNNs learn topology. We therefore define our practical downstream task as **synthetic graph augmentation for low-data GNN training**.

Diffusion models are promising for graph generation, but graphs are discrete, sparse, and non-Euclidean. *Unifying Generation and Prediction on Graphs with Latent Graph Diffusion* proposes encoding graphs into a latent space and running diffusion there, instead of directly over the raw adjacency matrix [Zhou et al. 2024]. However, it does not explicitly study which graph spectral frequency components are most useful for generation.

A very close prior work is *SPECTRE: Spectral Conditioning Helps to Overcome the Expressivity Limits of One-shot Graph Generators* [Martinkus et al. 2022]. SPECTRE conditions one-shot GAN graph generators on dominant Laplacian eigenvalues and eigenvectors, helping them control global graph structure and reduce mode collapse. However, SPECTRE is GAN-based, not diffusion-based. Thus, it does not answer whether spectral conditioning is useful inside latent graph diffusion.

Global and Local Topology-Aware Graph Generation via Dual Conditioning Diffusion shows that graph generation benefits from both local node-level and global cluster-level information [Xie and Pan 2026]. However, its global information is mainly cluster-based and does not directly analyze low-frequency versus high-frequency Laplacian components. *SDMG: Smoothing Your Diffusion Models for Powerful Graph Representation Learning* argues that low-frequency graph signals often capture meaningful global information, while reconstructing all details may over-emphasize high-frequency noise [Zhu et al. 2025]. SDMG motivates our idea, but it focuses on representation learning rather than graph generation.

Problem statement. Existing work shows that spectral information and global topology are useful, but it remains unclear whether **latent graph diffusion** benefits specifically from **low-frequency spectral conditioning**, and how the number of spectral components affects topology-only graph generation quality.

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2 Proposed Approach

Core idea. We propose to study a **frequency-aware latent graph diffusion** model. The model generates topology-only graphs in latent space, while the denoising process is conditioned on a small number of low-frequency graph Laplacian eigenvectors and eigenvalues. Low-frequency components describe the global skeleton of the graph, such as communities, bottlenecks, smooth structural regions, and large-scale organization. High-frequency components describe sharper local variations and may include noisy or less useful details.

For a graph $G = (V, E)$, we compute either the combinatorial graph Laplacian

$$L = D - A, \quad (1)$$

or the normalized graph Laplacian

$$L_{\text{norm}} = I - D^{-1/2}AD^{-1/2}, \quad (2)$$

where A is the adjacency matrix and D is the degree matrix. We then compute

$$Lu_i = \lambda_i u_i. \quad (3)$$

Eigenvectors with smaller eigenvalues are the low-frequency graph components.

Our method has four steps:

- (1) **Graph encoding:** encode each adjacency matrix into a latent representation using a graph autoencoder or a simplified latent diffusion setup.
- (2) **Spectral condition extraction:** compute the first k non-trivial low-frequency Laplacian eigenvectors/eigenvalues. For connected graphs, the first trivial eigenvector may be excluded.
- (3) **Conditional latent diffusion:** train a denoising model in latent space, conditioned on the spectral representation:

$$\epsilon_{\theta}(z_t, t, C_k), \quad (4)$$

where z_t is the noisy latent representation at diffusion step t , and C_k contains the selected spectral components.

- (4) **Graph decoding:** decode the generated latent representation back into an adjacency matrix.

Novelty. The novelty is not simply using Laplacian eigenvectors, since SPECTRE already uses spectral conditioning for GAN-based graph generation. Our contribution is to study spectral conditioning inside **latent graph diffusion**, and to systematically test which graph frequency bands are useful for diffusion-based generation. The key experiment is a **frequency cutoff gradient**: vary

$$k \in \{2, 4, 8, 16, 32\}, \quad (5)$$

and measure how generation quality changes. Comparing low-frequency conditioning against high-frequency and random spectral conditioning lets us test whether improvement comes specifically from meaningful low-frequency topology rather than simply from adding extra features.

3 Experimental Plan

Datasets. To keep the project focused, we will use two topology-only graph datasets:

- (1) **Planar graphs:** graphs with a clear global structural constraint. This tests whether spectral conditioning helps preserve global topology.
- (2) **Stochastic Block Model (SBM) graphs:** graphs with community structure. Low-frequency Laplacian eigenvectors are especially relevant for communities, so this directly tests our hypothesis.

Both dataset types are used in SPECTRE, making them suitable for comparison with prior graph generation literature [Martinkus et al. 2022].

Baselines and comparisons. We plan to compare:

- (1) **Unconditioned latent graph diffusion:** the same model without spectral conditioning.
- (2) **Low-frequency spectral conditioning:** our main method, using the first k non-trivial low-frequency Laplacian components.
- (3) **Frequency cutoff variants:** low-frequency conditioning with different values of k .
- (4) **High-frequency conditioning:** condition on high-frequency components instead of low-frequency components.
- (5) **Random spectral conditioning:** condition on randomly selected eigenvectors/eigenvalues.

Metrics. We will evaluate generated graph quality using a small subset of standard graph generation metrics, depending on implementation progress. Our primary metrics will be distributional similarity of graph statistics, especially degree distribution, clustering coefficient, and Laplacian spectrum MMD. When feasible, we may also report simple validity checks, such as whether generated planar graphs are planar and connected, or whether SBM graphs preserve community-like structure. If time allows, we will also report uniqueness and novelty.

Downstream utility experiment. As a practical use case, we will test whether generated graphs can help train a GNN in a low-data setting. We will train a GNN classifier to distinguish between planar and SBM-style graphs using: (i) a small real training set only, (ii) the small real set plus graphs from unconditioned diffusion, and (iii) the small real set plus graphs from our low-frequency spectral diffusion. If our generated graphs preserve useful topology, the GNN should generalize better to held-out real graphs.

Expected result. We expect that a few low-frequency components will improve global structural quality, while adding too many components may plateau or degrade performance. If low-frequency conditioning outperforms high-frequency or random conditioning, this supports the claim that diffusion benefits specifically from low-frequency global topology.

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